
Variant Impact Radar

Software Documentation

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A multi-page analytical workbench for exploring the predicted functional impact of human genetic variants, powered by the Ensembl Variant Effect Predictor (VEP) REST API.

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1. Introduction

Variant Impact Radar is a self-contained web application for exploring the predicted functional consequences of human genetic variants. Given a single variant identifier — most commonly a dbSNP rsID such as rs7903146 — the application queries the Ensembl Variant Effect Predictor (VEP) REST service, parses the returned annotation, computes a transparent heuristic impact score, and presents the evidence across a set of purpose-built pages: an at-a-glance overview, a detailed evidence breakdown, transcript-level consequences, a session-based comparison view, export tooling, and a full description of the scoring methodology.

The application is built with Streamlit and runs locally in a web browser. It requires no database, no account, and no configuration beyond a working internet connection to reach the Ensembl REST API.

1.1 Intended audience

The software is designed for students, educators, and researchers who want a fast, readable window into VEP annotation without writing code or navigating the full Ensembl web interface. It is equally suitable as a teaching aid for bioinformatics coursework and as a lightweight first-pass triage tool in research settings.

1.2 Design principles

- **Transparency.** Every number shown in the interface can be traced back to a visible rule. The scoring scheme is fully documented inside the application itself (Methodology page) and in Section 6 of this document.
- **Single source of truth.** All displayed values derive directly from the live Ensembl VEP response; the raw response is always inspectable and exportable.
- **Non-clinical by design.** The impact score is an evidence-aggregation heuristic intended for education and research triage. It is not a clinical pathogenicity classifier and must not be used for diagnosis, treatment, or patient management.

2. System Overview and Architecture

2.1 Processing pipeline

Each analysis follows a fixed five-stage pipeline, which the interface visualizes as the “Biological flow” strip on the Overview page:

Stage	Operation	Output
1. Variant	The identifier is validated and normalized.	A clean variant ID
2. Gene	The VEP response is parsed; the affected gene symbol is resolved and a gene-level summary is fetched from the Ensembl lookup endpoint.	Gene symbol and description
3. Transcript	One primary transcript is selected from all returned transcript consequences (see 5.3).	Selected transcript ID
4. Consequence	The most severe consequence and its annotation evidence (CADD, AlphaMissense) are extracted.	Primary consequence term
5. Evidence	The heuristic score is computed and mapped to an evidence tier.	Score (0–10) and tier

Table 1. The five-stage analysis pipeline.

2.2 Data sources

Service	Endpoint	Used for
Ensembl VEP	<code>rest.ensembl.org/vep/human/id/{id}</code>	Variant annotation: transcript consequences, colocated variants, population frequencies, clinical significance, CADD and AlphaMissense scores (requested via <code>CADD=1</code> and <code>AlphaMissense=1</code> parameters).
Ensembl Lookup	<code>rest.ensembl.org/lookup/symbol/homo_sapiens/{gene}</code>	Gene-level context: Ensembl gene ID, biotype, assembly, and free-text description.

Table 2. External REST services queried by the application.

Both requests use a 30-second timeout. VEP responses are cached in memory for 15 minutes and gene lookups for 60 minutes, so repeated queries within a session are served instantly and API load is kept low. HTTP 429 (rate limiting) and 404 (unknown identifier) responses are caught and reported as readable error messages.

2.3 Technology stack

- **Streamlit** (≥ 1.50) — application framework, multi-page navigation, widgets, and caching.
- **Plotly** — interactive gauge, radar, frequency, and comparison charts.
- **pandas** — tabular handling of transcript consequences, evidence rows, and population frequencies.
- **requests** — HTTP client for the Ensembl REST API.

3. Installation and Startup

3.1 Requirements

Python 3.10 or later and an internet connection. All Python dependencies are listed in `requirements.txt`.

3.2 Project layout

```
variant-impact-radar/
├── app.py                – the complete application
├── requirements.txt      – Python dependencies
├── README.md
├── assets/logo.svg      – sidebar wordmark
└── .streamlit/config.toml – color theme
```

The folder structure must be kept intact: the theme lives in `.streamlit/config.toml` and the sidebar logo in `assets/`.

3.3 Running the application

```
pip install -r requirements.txt
streamlit run app.py
```

Streamlit prints a local URL (by default `http://localhost:8501`) which opens in any modern browser. On first load the application automatically analyzes the built-in example variant `rs7903146` so every page

is populated immediately.

4. The Query Console

The sidebar is a persistent query console available on every page. It contains the variant input, the analyze action, curated example variants, the session history, a short description of the application, and a link to the Ensembl REST documentation.

4.1 Entering a variant

The input accepts dbSNP-style identifiers (rs numbers) as well as COSMIC and HGMD identifiers understood by the VEP `/vep/human/id` endpoint. Input is trimmed and validated against the character set `A-Z a-z 0-9 _ . : -`. Empty or malformed input produces an inline error without contacting the API.

4.2 Example variants

Four well-studied variants are provided as one-click examples:

Identifier	Gene	Common context
rs7903146	TCF7L2	Intronic variant; the strongest common genetic association with type 2 diabetes.
rs1801133	MTHFR	Missense variant (C677T) affecting folate metabolism.
rs1801131	MTHFR	Missense variant (A1298C), frequently studied alongside C677T.
rs1800795	IL6	Promoter-region variant studied in inflammatory phenotypes.

Table 3. Built-in example variants.

4.3 Session history

Every successfully analyzed variant is added to a session history rendered as selectable chips beneath the examples. Selecting a chip instantly reloads that variant's stored result without a new API call. The history also feeds the Compare variants page (Section 5.4).

4.4 Shareable links

The current variant is reflected in the browser URL as a query parameter, e.g. `?v=rs7903146`. Opening such a link starts the application and analyzes the referenced variant automatically, which makes analyses reproducible and easy to share. Deep links to specific pages (for example the Methodology page) are preserved and are not redirected.

5. Application Pages

The application is organized into six pages in three navigation groups. All pages share the query console and, where an analysis is loaded, a locus header showing the variant ID, chromosome and position, assembly, allele string, affected gene, retrieval timestamp, and the evidence tier.

Group	Page	Purpose
Workbench	Overview	At-a-glance result: summary cards, pipeline flow, score gauge, evidence radar, narrative interpretation.

Group	Page	Purpose
Workbench	Evidence & scoring	Variant metadata, full score breakdown, population allele frequencies, clinical annotations.
Workbench	Transcripts & gene	Selected transcript, filterable transcript consequence table, gene context, external links.
Tools	Compare variants	Side-by-side scores and evidence for variants analyzed in the session.
Tools	Reports & export	JSON, CSV, and HTML downloads plus the raw VEP response.
Reference	Methodology	The complete scoring rules, tier definitions, and scientific-use note.

Table 4. Page map.

5.1 Overview

The Overview page is the landing view after every analysis. It contains, top to bottom:

- **Locus header.** Variant identifier, genomic coordinates (chromosome, position, assembly), the allele string with each base rendered distinctly, the affected gene, the UTC retrieval time, and the evidence tier badge.
- **Summary cards.** Four metrics: affected gene, primary consequence (with the count of annotated terms), impact score out of 10, and evidence category.
- **Biological flow.** The five pipeline stages of Table 1 rendered as a left-to-right strip, each stage showing its resolved value for the current variant.
- **Impact gauge.** A 0–10 gauge with the tier boundaries marked as background bands and the current score as the needle value.
- **Evidence radar.** A four-axis radar normalizing the overall score, CADD (scaled from a 0–30 Phred range), AlphaMissense (scaled from its native 0–1 range), and the consequence weight onto a common 0–10 scale.
- **“Why this score?” popover.** An inline, expandable summary of exactly which rules contributed to the score, with a link to the full breakdown on the Evidence & scoring page.
- **Biological interpretation.** A generated plain-language paragraph describing the variant’s location, selected transcript, consequence class, available annotation evidence, gene description, and the resulting score and tier.

5.2 Evidence & scoring

This page decomposes the result into three tabs beneath a variant metadata section (chromosome, position, assembly, and alleles):

- **Score breakdown.** A table listing every evidence item — primary consequence, CADD-Phred, AlphaMissense, and each context bonus — with its observed value and its contribution to the total, alongside a bullet-point restatement of the applied scoring logic.
- **Population frequencies.** Allele frequencies extracted from the colocated-variant records of the VEP response, covering 1000 Genomes populations (global, African, American, East Asian, European, South Asian), gnomAD exomes and genomes cohorts, and ESP populations where available. Frequencies are shown as a horizontal bar chart and an exact-value table.
- **Clinical annotations.** Reported clinical significance terms (e.g. risk factor, likely risk allele), the minor allele and its frequency where reported, phenotype descriptions, and linked PubMed identifiers

(the first eight are shown as chips, with the total count stated).

The population-frequency and clinical tabs display whatever the VEP response contains; for variants with no colocated records the tabs state that no data was returned rather than showing empty charts.

5.3 Transcripts & gene

Selected transcript. The application selects one primary transcript per analysis, preferring canonical transcripts first, then MANE-annotated transcripts, then the transcript with the richest annotation detail. The selection is shown with its canonical and MANE status and, for coding variants, the protein change.

Transcript consequences. Every transcript-level annotation in the VEP response is listed in a table with transcript ID, gene, consequence terms, canonical and MANE flags, amino-acid change, and codon change. A segmented control filters the table to All, Canonical only, or MANE only, with a live count of displayed rows.

Gene context. The Ensembl gene ID, biotype, species, assembly, and free-text gene description from the lookup service.

External evidence links. One-click links to the Ensembl variant page, an NCBI dbSNP search for the identifier, and the Ensembl gene page.

5.4 Compare variants

All variants analyzed during the session can be compared side by side. A multi-select chooses which variants to include; the page then shows a comparison table (variant, gene, primary consequence, impact score rendered as a progress bar, evidence tier, CADD-Phred, AlphaMissense), a tier-colored bar chart of impact scores, and a multi-trace radar overlaying each variant's evidence profile. The page states explicitly that comparison uses the same transparent heuristic and does not imply clinical ranking.

5.5 Reports & export

Three portable artifacts can be downloaded from the current analysis; see Section 7 for exact contents and file names. The page also contains an expandable view of the complete, unmodified VEP response for verification and debugging.

5.6 Methodology

A reference page reproducing, inside the application, everything in Section 6 of this document: the consequence weight table, annotation thresholds, context bonuses, tier legend, cache durations, and the scientific-use note.

6. Scoring Methodology

The impact score is an additive heuristic on a 0–10 scale. It combines a base weight for the primary consequence class with fixed adjustments for annotation evidence and context. The total is clamped to the 0–10 range and mapped to an evidence tier. The scheme is intentionally simple and fully visible; it is designed to be explainable, not to compete with trained pathogenicity classifiers.

6.1 Base consequence weights

Consequence class	Weight
transcript_ablation	10
splice_acceptor_variant	9
splice_donor_variant	9
stop_gained	9
frameshift_variant	9
start_lost	8
stop_lost	8
missense_variant	6
inframe_insertion	5
inframe_deletion	5
protein_altering_variant	5
regulatory_region_variant	3
TF_binding_site_variant	3
promoter_variant	3
5_prime_UTR_variant	2
3_prime_UTR_variant	2
non_coding_transcript_exon_variant	2
intron_variant	1
synonymous_variant	1
intergenic_variant	0

Table 5. Base weights per consequence class. Consequence terms not in this table contribute a default low weight.

6.2 Annotation evidence thresholds

Evidence	Condition	Adjustment
CADD-Phred	≥ 30	+2.0
CADD-Phred	≥ 20	+1.0
CADD-Phred	≥ 10	+0.5
AlphaMissense	≥ 0.90	+1.5
AlphaMissense	≥ 0.70	+1.0
AlphaMissense	≥ 0.50	+0.5

Table 6. Threshold adjustments. Only the highest matching row per evidence type is applied. Scores are read from the selected transcript's annotation; when a value is absent from the response, no adjustment is made and the interface reports "Not available".

6.3 Context bonuses

Context signal	Adjustment
Selected transcript is canonical	+0.5
Selected transcript carries a MANE annotation	+0.5
Known colocated-variant context is present	+0.5
Phenotype annotation is present	+0.5
Gene-level Ensembl context was retrieved	+0.25

Table 7. Fixed context bonuses.

6.4 Evidence tiers

Tier	Condition	Reading
HIGH	score $\geq$ 8	Potentially high-impact computational evidence
MODERATE	score $\geq$ 5	Moderate computational impact
LOW-MODERATE	score $\geq$ 2.5	Limited computational impact
LOW	score $<$ 2.5	Low computational impact

Table 8. Mapping from clamped score to evidence tier.

7. Data Exports

Format	File name	Contents
JSON	{variant}_variant_report.json	The full machine-readable payload: variant metadata, selected transcript, consequence summary, score with itemized reasons, interpretation text, gene summary, and the raw VEP response.
CSV	{variant}_transcripts.csv	The complete transcript consequence table, one row per transcript annotation.
HTML	{variant}_variant_report.html	A self-contained, printable report of the analysis suitable for sharing or archiving. Opens in any browser; no external assets required.

Table 9. Export artifacts. {variant} is replaced by the analyzed identifier.

In addition, the raw Ensembl VEP response is viewable on the Reports & export page and embedded in the JSON export, so any displayed value can be audited against its source.

8. Use Cases

8.1 Teaching and coursework

The application makes an abstract pipeline — identifier, annotation service, consequence ontology, scoring, interpretation — concrete and inspectable. Instructors can walk students through the Biological flow strip, then open the Methodology page to connect each displayed number to a rule. The example variants cover an intronic association signal (rs7903146), two classic missense variants (rs1801133, rs1801131), and a promoter variant (rs1800795), giving contrasting consequence classes out of the box.

8.2 First-pass research triage

For small candidate lists (for example, variants highlighted by an association study or a reading of the literature), the workbench provides rapid orientation: consequence class, canonical-transcript effect, deleteriousness scores where available, population frequencies, and any reported clinical significance — all in one place, with links out to Ensembl and dbSNP for deeper review. The Compare variants page supports quick relative ranking of a handful of candidates within a session.

8.3 Reproducible sharing and journal clubs

Because analyses are addressable by URL (?v=rsID) and exportable as self-contained HTML reports, a specific view of a variant can be shared with colleagues or attached to notes. The JSON export preserves the raw VEP response for full reproducibility, including later re-verification against updated Ensembl releases.

8.4 Demonstrating API-driven application design

The project doubles as a worked example of a clean REST-API-driven analytical application: input validation, cached HTTP calls, defensive parsing of a complex nested response, transparent derived metrics, and multi-page presentation. The single-file implementation is deliberately readable for this purpose.

9. Limitations and Responsible Use

- **Not a clinical tool.** The impact score is a fixed-rule heuristic for education and triage. It is not validated for clinical use and must not inform diagnosis, treatment, or patient management.
- **Annotation availability varies.** CADD and AlphaMissense values are only present for a subset of variants and transcripts; absent values contribute nothing to the score and are reported as “Not available”.
- **Single-variant scope.** The application annotates one identifier per query. It does not process VCF files, genomic coordinates, or batch input.
- **Dependence on a live service.** Results reflect the current Ensembl release at query time and require network access. Ensembl rate limits (HTTP 429) are surfaced as readable errors.
- **Session-scoped history.** The history and comparison features cover the current browser session; closing the application clears them. Durable records should be kept via the export formats.

10. Troubleshooting

Symptom	Likely cause and resolution
“Variant ID contains unsupported characters.”	The input contains characters outside A–Z, a–z, 0–9, underscore, dot, colon, or hyphen. Re-enter the identifier exactly as published (e.g. rs7903146).
“Variant not found” / HTTP 404	The identifier is not known to the VEP id endpoint. Verify the rsID in dbSNP, or try the current merged identifier if the rsID was retired.
Rate-limit error / HTTP 429	The Ensembl REST service is throttling requests. Wait a short time and retry; cached variants remain available from the session history.
Request timeout	The Ensembl service is slow or unreachable. The application waits up to 30 seconds per request; check connectivity and retry.
Population frequencies or clinical tab is empty	The VEP response contained no colocated-variant records for this identifier. This is a property of the data, not an error.
Theme or logo missing	The application was started outside its folder. Run streamlit from the project directory so .streamlit/config.toml and assets/ are found.

Table 10. Common issues and resolutions.

11. Glossary

Term	Definition
rsID	A Reference SNP identifier assigned by dbSNP, e.g. rs7903146.
VEP	The Ensembl Variant Effect Predictor, which annotates the functional consequences of variants against Ensembl transcripts.
Consequence	A Sequence Ontology term describing a variant's predicted effect on a transcript (e.g. missense_variant, intron_variant).
Canonical transcript	The single representative transcript Ensembl designates for a gene.
MANE	Matched Annotation from NCBI and EMBL-EBI: transcripts on which the two annotation systems agree.
CADD (Phred)	Combined Annotation Dependent Depletion; a scaled deleteriousness score where higher values indicate variants predicted to be more deleterious (≥ 20 corresponds roughly to the top 1% of possible substitutions).
AlphaMissense	A DeepMind model scoring missense variants from 0 to 1 by likelihood of pathogenicity.
gnomAD	The Genome Aggregation Database; large-scale population allele-frequency reference (exomes and genomes cohorts).
1000G	The 1000 Genomes Project; continental-population allele frequencies.
MAF	Minor allele frequency: the frequency of the less common allele in a population.
PMID	A PubMed identifier referencing a published article linked to the variant.
Colocated variant	A known variant record at the same genomic position in the VEP response; the source of frequency, clinical, and literature annotations.

Table 11. Terms used throughout the application.